Deviance Explained

Logistic Regression

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Example

```
options(contrasts = c("contr.treatment", "contr.poly"))
ldose <- rep(0:5, 2)
numdead <- c(1, 4, 9, 13, 18, 20, 0, 2, 6, 10, 12, 16)
sex <- factor(rep(c("M", "F"), c(6, 6)))
SF <- cbind(numdead, numalive = 20 - numdead)

budworm.lg <- glm(SF ~ sex*ldose, family = binomial)
summary(budworm.lg, cor = F)
```

R output

```
##
## Call:
## glm(formula = SF ~ sex * ldose, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.9935     0.5527  -5.416 6.09e-08 ***
## sexM          0.1750     0.7783   0.225    0.822
## ldose         0.9060     0.1671   5.422 5.89e-08 ***
## sexM:ldose    0.3529     0.2700   1.307    0.191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 124.8756  on 11  degrees of freedom
## Residual deviance:   4.9937  on  8  degrees of freedom
## AIC: 43.104
##
## Number of Fisher Scoring iterations: 4
```

Deviances are stored in `summary` object

```
summaryLog <- summary(budworm.lg)
# 'Residual deviance' is the deviance;
# 'Null deviance' is the null deviance

(summaryLog$deviance)
```

```
## [1] 4.993727
```

```
(summaryLog$null.deviance)
```

```
## [1] 124.8756
```

- Recall: Intuitively, if the null deviance is much larger than the fitted deviance, the predictors are important, if they are similar the predictors are noise.

Fit the null model to check

```r
# Null model - only intercept
budworm.null <- glm(SF ~ 1, family = binomial)
summary(budworm.null, cor = F)
```

```
##
## Call:
## glm(formula = SF ~ 1, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.1503     0.1295  -1.161    0.246
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 124.88  on 11  degrees of freedom
## Residual deviance: 124.88  on 11  degrees of freedom
## AIC: 156.99
##
## Number of Fisher Scoring iterations: 3
```

Calculate R^2

Recall:

$$R^2 = 1 - \frac{D}{D_0} = 1 - \frac{\text{deviance of fitted model to saturated}}{\text{deviance of null model to saturated}}.$$

- It is a quantity between 0 and 1.
- If the fit is perfect, then $D = 0$ and $R^2 = 1$, but if the predictors have no information about Y, then $D = D_0$, and $R^2 = 0$.
- For the example at hand,

```
(R_sq = 1 - summaryLog$deviance / summaryLog$null.deviance)
```

```
## [1] 0.9600104
```

- Is the high value of R^2 surprising or not?

Smarket

```r
library(ISLR)
names(Smarket)
```

```
## [1] "Year"      "Lag1"      "Lag2"      "Lag3"      "Lag4"      "Lag5"
## [7] "Volume"    "Today"     "Direction"
```

```r
dim(Smarket)
```

```
## [1] 1250    9
```

Smarket data

```
glm.fits=glm(Direction~Lag1+Lag2+Lag3+Lag4+Lag5+Volume,data=Smarket,f
summary(glm.fits)
```

```
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##     Volume, family = binomial, data = Smarket)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.126000   0.240736  -0.523    0.601
## Lag1        -0.073074   0.050167  -1.457    0.145
## Lag2        -0.042301   0.050086  -0.845    0.398
## Lag3         0.011085   0.049939   0.222    0.824
## Lag4         0.009359   0.049974   0.187    0.851
## Lag5         0.010313   0.049511   0.208    0.835
## Volume       0.135441   0.158360   0.855    0.392
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1731.2  on 1249  degrees of freedom
## Residual deviance: 1727.6  on 1243  degrees of freedom
## AIC: 1741.6
```

Smarket data

- Calculate the R^2 for this fitted model.

```
summarySmarket <- summary(glm.fits)
(R_sq = 1 - (summarySmarket$deviance / summarySmarket$null.deviance))
```

```
## [1] 0.002074126
```

- Now, what does this say about the quality of fit?
- How do you connect this with the p-values?